

SARA BAKHTIARI

PhD Candidate in Mechanical Engineering | Scientific Machine Learning | Computational Modeling | Multimodal AI

University of Kentucky · Lexington, KY | Sara.b@uky.edu | [LinkedIn](#) | [GitHub](#) | [ResearchGate](#) | [Website](#)

Python · Vision-Language Models · Embeddings · Regression · Scientific Computing · HPC · Computational Biomechanics

PROFESSIONAL SUMMARY

PhD candidate in Mechanical Engineering at the University of Kentucky working across scientific computing, computational biomechanics, and machine learning. Develops physics-based finite-element simulations and data-driven workflows, including vision-language-model embeddings, learned bottleneck representations, regression, repeatability analysis, and cross-validation. Experienced with Python, HPC, large scientific datasets, numerical modeling, and translating complex physical systems into robust computational pipelines.

EDUCATION

University of Kentucky — Lexington, KY

Spring 2024–Present

PhD in Mechanical Engineering, Direct PhD Program · PhD Candidate since April 2026 · GPA: 3.91/4.00 · Advisor: Jonathan F. Wenk, PhD

Iran University of Science and Technology (IUST) — Tehran, Iran

2016–2021

B.Sc. in Mechanical Engineering

RESEARCH EXPERIENCE

Graduate Research Assistant — Computational Cardiac Mechanics

Spring 2024–Present

University of Kentucky · Advisor: Jonathan F. Wenk, PhD

- Develop large-scale finite-element cardiac simulations in Python/FEniCS that couple mechanics, active contraction, circulation, and spatial biological state variables.
- Design reproducible HPC workflows using MPI, PETSc, SLURM, structured simulation outputs, and automated post-processing for high-dimensional spatial/time-series data.
- Developed computational studies of constitutive uncertainty, fiber disarray, mechanical efficiency, and physiological compensation, integrating model outputs across local and organ scales.
- Currently coupling an agent-based fibrosis model with finite-element mechanics, creating a multiscale computational framework linking spatial biological evolution to tissue mechanics.

Vision-Language Models & Machine Learning for Electrospinning

Additional research project

- Applied pretrained and fine-tuned vision-language models to electrospinning videos and extracted high-dimensional hidden-state embeddings for downstream prediction.
- Developed learned low-dimensional bottleneck representations and compared them with full embeddings and randomly selected features.
- Designed repeated embedding-generation, feature-selection, and cross-validation studies to quantify representation variability and predictive robustness.
- Built regression pipelines for predicting electrospun fiber diameter from video-derived representations using Python and scikit-learn.

PUBLICATIONS & RESEARCH OUTPUTS

Islam, R., Bakhtiari, S., Anugrah, G. G., Mehri, M., & Wenk, J. F. “Advancements in finite element modeling of myocardial infarction: a review.” *Biomechanics and Modeling in Mechanobiology*, 25(4), 2026. DOI: 10.1007/s10237-026-02107-0

Manuscripts Under Review

- Bakhtiari, S., Mehri, M., & Wenk, J. F. “Choice of Passive Myocardial Constitutive Law Drives Divergent Predictions of Left Ventricular Remodeling in Volume-Overload Conditions.” Manuscript under review.
- Bakhtiari, S., Mehri, M., & Wenk, J. F. “Effect of Myocardial Fiber Disarray on Left Ventricular Mechanical Efficiency and Baroreflex Compensation.” Manuscript under review.

Selected Conference Presentation

Bakhtiari, S., Mehri, M., & Wenk, J. F. “Choice of Passive Myocardial Constitutive Law Drives Differences in LV Mechanics and Growth Under Disease Conditions.” Poster, Biomedical Engineering Society (BMES) Annual Meeting, San Diego, CA, October 2025.

TEACHING EXPERIENCE

Teaching Assistant — Statics, University of Kentucky

Fall 2026

Supported undergraduate instruction and problem solving in equilibrium, force/moment analysis, trusses, and frames.

Teaching Assistant — Statics, Iran University of Science and Technology

2021

TECHNICAL SKILLS

Machine Learning: scikit-learn, regression, cross-validation, feature selection, embeddings, dimensionality reduction, learned bottlenecks, vision-language models

Programming & Data: Python, NumPy, SciPy, Pandas, MATLAB, structured scientific data processing

Scientific Computing: FEniCS/DOLFIN, UFL, nonlinear numerical simulation, model validation

HPC: MPI, mpi4py, PETSc, SLURM, Linux, distributed workflows, Singularity/container environments

Domain Modeling: Computational biomechanics, cardiovascular mechanics, constitutive modeling, multiscale fibrosis modeling